# BUSINESS REQUIREMENTS DOCUMENT (BRD)

## *E- COMMERCE ORDER MANAGEMENT SYSTEM (OMS)*

**Prepared By:**

**Revathi Kamindla**

Business Analyst (Project)

April 2026 | Version 1.0

## Document Control

| Version | Date | Author | Description |
|---|---|---|---|
| **0.1** | April 2026 | Revathi Kamindla | Initial Draft |
| **1.0** | April 2026 | Revathi Kamindla | Final Version for Review |

## Approval Sign-Off

| Role | Name | Signature | Date |
|---|---|---|---|
| **Business Owner** | | | |
| **Project Manager** | | | |
| **Business Analyst** | Revathi Kamindla | | April 2026 |

## 1. Executive Summary

The E-Commerce Order Management System (OMS) is a proposed centralized platform designed to streamline the end-to-end order lifecycle for an online retail business. The current order management process is fragmented, heavily manual, and lacks real-time visibility — resulting in delayed order processing, customer dissatisfaction, and operational inefficiencies.

This Business Requirements Document (BRD) outlines the high-level business needs, objectives, scope, and stakeholder expectations for the proposed OMS. The document serves as the foundation for all subsequent analysis, design, and development activities related to this project.

The OMS will enable customers to place, track, cancel, and return orders through a centralized system while providing administrators with a dashboard to manage and monitor all order activities efficiently.

## 2. Business Background

The rapid growth of e-commerce has significantly increased the complexity of managing customer orders. Businesses must coordinate across multiple functions including order placement, payment processing, packaging, shipment, and returns — all while maintaining accuracy and speed.

In the current environment, organizations rely on fragmented systems and manual processes to manage these activities. This results in:

- Delayed order processing and fulfillment
- Inaccurate or outdated order tracking information
- Poor coordination between customers, administrators, vendors, and delivery partners
- High dependency on manual intervention leading to errors
- Reduced customer satisfaction and trust

A centralized Order Management System is required to address these challenges and support the organization's growth objectives.

## 3. Problem Statement

| Problem Area | Current Challenge | Business Impact |
|---|---|---|
| **Order Tracking** | No real-time tracking available to customers | Customer dissatisfaction and increased support calls |
| **Order Status Updates** | Manual email-based updates by admin | Delayed and inconsistent communication |
| **Order Cancellation** | Customers must call or email to cancel | Slow processing, no immediate confirmation |
| **Return and Refund** | Manual paperwork-based process | Lengthy resolution time, poor visibility |
| **Admin Monitoring** | Spreadsheet-based order management | Error-prone, lacks real-time visibility |
| **Stakeholder Coordination** | Phone and email-based communication | Fragmented, delayed, and unreliable |

## 4. Business Objectives

| Objective ID | Business Objective | Success Metric |
|---|---|---|
| **BO-01** | Improve overall order processing efficiency | Reduction in order processing time by 40% |
| **BO-02** | Enhance customer satisfaction through real-time visibility | Customer satisfaction score improvement |
| **BO-03** | Reduce manual intervention and operational errors | 60% reduction in manual order handling |
| **BO-04** | Ensure better stakeholder coordination | Faster communication across all parties |
| **BO-05** | Increase process transparency across order lifecycle | 100% order status visibility in real time |
| **BO-06** | Support business growth through scalable framework | System supports 30% annual growth in orders |

## 5. Project Scope

### 5.1 In Scope

| # | Feature |
|---|---|
| 1 | User registration, login, and account management |
| 2 | Product browsing and selection |
| 3 | Order placement and confirmation |
| 4 | Payment status tracking (Success, Pending, Failed) |
| 5 | Real-time order tracking and status updates |
| 6 | Order lifecycle management (Placed → Processed → Shipped → Delivered) |
| 7 | Order cancellation functionality |
| 8 | Returns and refund initiation process |
| 9 | Administrative dashboard for order management |
| 10 | System notifications for order status updates |

### 5.2 Out of Scope

| # | Excluded Feature | Reason |
|---|---|---|
| 1 | Inventory and warehouse management | Separate system scope |
| 2 | Third-party logistics integration | High-level assumption only |
| 3 | Supplier/vendor management system | Out of current project boundary |
| 4 | Advanced analytics and reporting | Future phase |
| 5 | AI-based product recommendations | Future phase |
| 6 | Mobile application development | Platform-specific, out of scope |
| 7 | Payment gateway internal processing | External system responsibility |

## 6. Stakeholder Analysis

### 6.1 Stakeholder Register

| Stakeholder | Role | Interest | Influence | Expectation |
|---|---|---|---|---|
| **Customers** | End Users | High | Medium | Seamless ordering and real-time tracking |
| **System Administrators** | Internal Users | High | High | Centralized order management dashboard |
| **Vendors / Sellers** | Internal/External | High | Medium | Clear order fulfillment instructions |
| **Delivery Partners** | External | Medium | Low | Accurate shipment details |
| **Payment Gateway Providers** | External | Low | Low | Secure payment processing |
| **Business Owners** | Business Stakeholders | High | High | Improved efficiency and customer satisfaction |
| **Business Analyst** | Project Team | High | High | Clear requirements and approvals |

### 6.2 RACI Matrix

| Activity | Customer | Admin | Vendor | Delivery Partner | BA | Business Owner |
|---|---|---|---|---|---|---|
| **Place Order** | R | I | I | — | — | — |
| **Process Order** | I | R | A | — | — | — |
| **Pack & Ship** | — | I | R | A | — | — |
| **Update Status** | — | R | I | I | — | — |
| **Cancel Order** | R | A | I | — | — | — |
| **Return/Refund** | R | A | I | — | — | C |
| **Requirements Sign-Off** | — | C | — | — | R | A |

## 7. Business Requirements

### 7.1 Functional Business Requirements

| Req ID | Business Requirement | Priority | MoSCoW |
|---|---|---|---|
| **BR-01** | System shall allow customers to register and log in securely | High | Must Have |
| **BR-02** | System shall allow customers to browse products and place orders | High | Must Have |
| **BR-03** | System shall generate a unique Order ID for each confirmed order | High | Must Have |
| **BR-04** | System shall provide real-time order status updates | High | Must Have |
| **BR-05** | System shall allow customers to cancel orders before shipment | High | Must Have |
| **BR-06** | System shall support return and refund initiation | Medium | Should Have |
| **BR-07** | System shall allow administrators to manage and update order statuses | High | Must Have |
| **BR-08** | System shall notify customers of key order updates | Medium | Should Have |
| **BR-09** | System shall provide admin dashboard for order activity overview | Medium | Should Have |
| **BR-10** | System shall support multiple payment methods | Low | Could Have |

### 7.2 Non-Functional Business Requirements

| Req ID | Business Requirement | Standard | MoSCoW |
|---|---|---|---|
| **NFR-01** | System page load time | ≤ 3 seconds | Must Have |
| **NFR-02** | Concurrent user support | 10,000 simultaneous users | Must Have |
| **NFR-03** | System uptime | 99.9% SLA | Must Have |
| **NFR-04** | Data encryption | AES-256 or equivalent | Must Have |
| **NFR-05** | Recovery Time Objective | RTO ≤ 15 minutes | Should Have |
| **NFR-06** | Recovery Point Objective | RPO ≤ 1 hour | Should Have |
| **NFR-07** | Order volume scalability | 1,00,000 orders/day at peak | Should Have |

## 8. Assumptions

| # | Assumption |
|---|---|
| 1 | Users have access to stable internet connectivity and basic digital literacy |
| 2 | Payment gateways function correctly and securely |
| 3 | Delivery partners provide accurate and timely order status updates |
| 4 | Stakeholders will actively participate in requirement gathering and validation |
| 5 | Data used for analysis is accurate and consistent |
| 6 | The system operates in a secure and controlled environment |
| 7 | All stakeholders will provide timely sign-off on documented requirements |

## 9. Constraints

| # | Constraint | Impact |
|---|---|---|
| **1** | Limited project timeline for analysis and documentation | Scope must remain focused |
| **2** | No actual system development involved (documentation only) | Analysis is conceptual |
| **3** | External integrations are assumed and not deeply integrated | High-level analysis only |
| **4** | Scope restricted to order management processes only | Full e-commerce platform excluded |
| **5** | Limited access to real-time business data | Hypothetical data used |

## 10. Risk Register

| Risk ID | Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|---|
| **R-01** | Payment gateway downtime | Medium | High | Backup payment provider |
| **R-02** | Data breach or unauthorized access | Low | High | Encryption and RBAC |
| **R-03** | High traffic performance degradation | Medium | High | Load balancing |
| **R-04** | Incorrect order status updates | Medium | Medium | Automated validation rules |
| **R-05** | Delivery partner API failure | Medium | High | Admin manual override |
| **R-06** | Scope creep | High | Medium | Strict change control process |
| **R-07** | Stakeholder unavailability | Low | Medium | Advance scheduling |

## 11. Dependencies

| # | Dependency | Type | Impact if Unavailable |
|---|---|---|---|
| **1** | Payment gateway availability | External | Orders cannot be processed |
| **2** | Delivery partner system | External | Shipment tracking unavailable |
| **3** | Stakeholder sign-off on requirements | Internal | Development cannot begin |
| **4** | Admin user availability for testing | Internal | UAT cannot be completed |

## 12. Approval

| Role | Name | Approval Status | Date |
|---|---|---|---|
| **Business Owner** | | Pending | |
| **Project Manager** | | Pending | |
| **Business Analyst** | Revathi Kamindla | Approved | April 2026 |